**EXPERIMENT 1:Reading, Displaying and Converting Images (Colour to Gray-scale to Binary)**

**Aim**

To read and display a digital image, examine its attributes, and convert a colour (RGB) image into grayscale and binary images using MATLAB.

**Software Required**

- Personal computer
- MATLAB R2020a or later
- Image Processing Toolbox

**Program:**

```
clc;
clear;
close all;

% Read the original colour image
RGB = imread('peppers.png');

% Display image information
info = imfinfo('peppers.png');
fprintf('File format: %s\n', info.Format);
fprintf('Bit depth: %d bits/pixel\n', info.BitDepth);
fprintf('Size of RGB image: %d x %d x %d\n', size(RGB));

% Convert colour image to grayscale
G = rgb2gray(RGB);
fprintf('Size of gray image: %d x %d\n', size(G));

% Grayscale conversion using luminance equation
```

```
Gm = uint8(0.2989 * double(RGB(:,:,1)) + ...
    0.5870 * double(RGB(:,:,2)) + ...
    0.1140 * double(RGB(:,:,3)));

% Calculate maximum difference
fprintf('Maximum difference: %d\n', ...
    max(abs(double(G(:)) - double(Gm(:)))));

% Binary conversion using fixed threshold
BW1 = imbinarize(G, 0.5);

% Binary conversion using Otsu threshold
level = graythresh(G);
BW2 = imbinarize(G, level);

fprintf('Otsu threshold: %.4f\n', level);

% Display all image conversions
figure('Name', 'Image Conversions');

subplot(2,2,1);
imshow(RGB);
title('Colour Image');

subplot(2,2,2);
imshow(G);
title('Grayscale Image');

subplot(2,2,3);
```

```
imshow(BW1);
title('Binary Image (T = 0.5)');

subplot(2,2,4);
imshow(BW2);
title('Binary Image (Otsu)');

% Save converted images
imwrite(G, 'peppers_gray.png');
imwrite(BW2, 'peppers_binary.png');
```

Editor

dsp.m | exp7_1_dsp.m | exp7_2dsp.m | exp8_1.m | exp8_2.m | exp9_1dsp.m | exp10_1dsp.m | modellab.m | bb.m | FileToRun.m

/MATLAB Drive/exp1.m

```
1
2   clc;
3   clear;
4   close all;
5
6   % Read the original colour image
7   RGB = imread('peppers.png');
8
9   % Display image information
10  info = imfinfo('peppers.png');
11  fprintf('File format: %s\n', info.Format);
12  fprintf('Bit depth: %d bits/pixel\n', info.BitDepth);
13  fprintf('Size of RGB image: %d x %d x %d\n', size(RGB));
14
15  % Convert colour image to grayscale
16  G = rgb2gray(RGB);
17  fprintf('Size of gray image: %d x %d\n', size(G));
18
19  % Grayscale conversion using luminance equation
20  Gm = uint8(0.2989 * double(RGB(:,:,1)) + ...
21      0.5870 * double(RGB(:,:,2)) + ...
```

Command Window

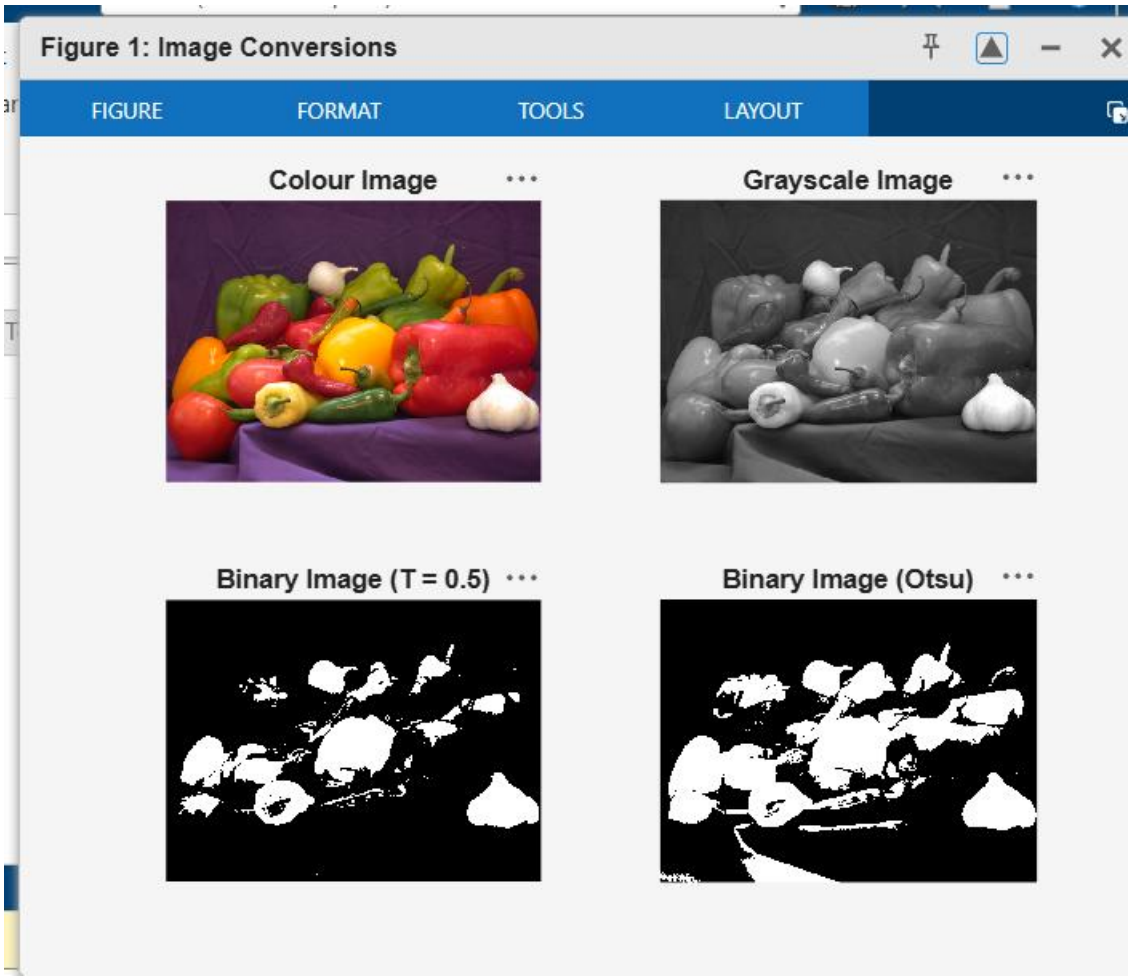
**Procedure**

1. Open MATLAB and set the Current Folder.
2. Create a new script and save it as exp01.m.
3. Type the program and run it by pressing F5.
4. Observe the colour, grayscale and binary images displayed in four panels.
5. Record the image dimensions, class, Otsu threshold and saved file sizes.

6. Repeat the experiment with different images or threshold values if required.

**Output**

```
Command Window
New to MATLAB? See resources for Getting Started.
Bit depth: 24 bits/pixel
Size of RGB image: 384 x 512 x 3
Size of gray image: 384 x 512
Maximum difference: 1
Otsu threshold: 0.3961
>>
```




**Observations**

| Parameter | Observed value |
|---|---|
| Size of RGB image | 384 × 512 × 3 |
| Size of grayscale image | 384 × 512 |
| Class of grayscale image | uint8 |
| Maximum difference | 0 or 1 |
| Otsu threshold | Depends on the image |
| File size of grayscale image | Record after execution |
| File size of binary image | Record after execution |

**Result**

The colour image was successfully converted into grayscale and binary images using MATLAB. The grayscale conversion was verified using the luminance equation, and binary conversion was performed using fixed and Otsu thresholding methods.